Flight Dynamics of the Bell Boeing V-22 Osprey in JSBSim

A blade-element tiltrotor model, built and checked as the XV-15 from NASA's data and carried to the V-22 with its published rotor, wing and performance data

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Abstract

A flight-dynamics model of the Bell Boeing V-22 Osprey for the FlightGear simulator has been written in JSBSim, replacing the parametric YASim model the aircraft used before. The V-22's engineering data are not public; those of its technology demonstrator, the Bell XV-15, are: Bell's real-time simulation data from its wind tunnel tests, NASA's hover test of its full-scale rotor and its flight tests. The model is therefore built and checked as an XV-15 first and carried to the V-22 with the V-22's published rotor geometry, wing, weights, engine ratings and performance charts. Each of the two three-bladed gimballed proprotors is simulated blade by blade, eight segments per blade in cambered-section tables at their own angle of attack and Mach number, each blade twisting in its first torsion mode, under a Peters–He finite-state wake with ground effect; the gimbal's two degrees of freedom carry the pitch–flap coupling ($\delta_3 = -15°$). The nacelles tilt from 97.5° to airplane mode inside the conversion corridor; Bell's wing, fuselage and empennage tables, scaled, sit in the rotors' ring-vortex wakes; an interconnected drive with its torque limit, two engines, the collective governor and a stability augmentation complete it. The whole model, 622 aerodynamic functions and 744 control-system components, is generated by a script from transcribed data and runs in plain JSBSim at 480 Hz. As an XV-15 its isolated rotor needs −2.8 to +3.4 % of the measured hover power from $C_T/\sigma = 0.08$ to 0.18 (the sections' stall and minimum drag fitted) and reaches $C_T/\sigma = 0.184$ against the test's 0.181; its wing's download follows the flight tests' within 1.2 point of thrust (fitted). In level flight, with the test altitudes unpublished, with the drag Ferguson added from them it needs −27 to −16 % of the measured power at 10 000 ft in airplane mode with the gear up, while through the conversion it needs more than the test at both altitudes. As a V-22, with 11.5 ft^2 of parasite drag fitted to Boeing's airframe polar and the drive's torque limit fitted to the sea-level hover ceiling, its maximum level speed at 42 000 lb is 293 and 319 KTAS at sea level and 17 000 ft against 274 and 321, its hover ceiling at 8400 and 16 000 ft −2 and +5 % from Boeing's chart, and its pitch attitude through the conversion within 3.3° (rms) of the flight test's. Deficiencies are quantified: the XV-15's power in helicopter mode, +24 % at 45 kt and −8 % at 94 kt; its collective gradient in hover, 20 % steeper than measured; the V-22 wing's download, 11.8 % of thrust against 9.2; and the wing's share of the lift at 30° of nacelle. In descent each rotor's inflow follows Johnson's vortex ring state model: in vertical descent at a fixed lever the V-22 runs away between 1933 and 7147 ft/min, and in his encounter it drops and then rolls off, each rotor on its own velocities.

1 Introduction

The V-22 distributed for FlightGear flies on YASim, whose helicopter model builds a rotor from a short list of geometric parameters and tuning factors and has no notion of a tiltrotor: the stock aircraft tilts its rotors' thrust and lets the wing take over. What makes the tiltrotor hard is precisely what such a model leaves out: two large, highly twisted proprotors that have to hover like a helicopter's rotor and cruise like propellers, the gimbal that lets each disc tilt as a whole, the wing that sits in their downwash in the hover (its download) and in their slipstream in cruise, and the conversion between the two, flown inside a corridor of nacelle angle against airspeed.

The V-22's own engineering data are not public. Its technology demonstrator's are: Bell documented the XV-15's real-time simulation in 1973 down to every wind tunnel table [1], NASA tested its full-scale rotor in hover [3], and NASA and the Army flew it and compared the simulation with the flights [4, 5]. For the V-22 itself Boeing and Bell published the rotor's geometry, the wing's planform, the airframe's lift and drag, the download, hover ceilings, the flight envelope [6] and the flight test results through the conversion [7]; NASA the rotor's characteristics [8, 9]. So the model is built as an XV-15 first, checked against the XV-15's tests, and then carried to the V-22 by its published data, each XV-15 quantity that has no V-22 counterpart scaled by geometry and marked as such.

This report describes the JSBSim [33] model built on that material, after UH-60A and Bo 105 models built the same way, and how well it reproduces the tests. All of the flight model, the blades' integration, the gimbals, the wakes, the engines and drive, the governor and the augmentation, is JSBSim XML; the FlightGear side only animates and instruments it. All result tables, figures and the numbers quoted from them are written by the evaluation scripts.

2 Reference data

Table 1 lists the sources; most are scans, and every curve used was digitised from them. A few points of data reduction matter for the comparisons. *(i)* Bell's simulation data [1] are complete but for three values illegible in the scan. The wing–pylon pitching moment at zero lift for the third flap setting (page B-12) is taken as −0.20 (Ferguson's revision [26] has −0.19 in airplane mode). A pylon's moment of inertia about its shaft, 81.4 or 31.4 slug ft^2, is 81.4: with it the aircraft's inertia changes with the tilt as Ferguson's does within 4 %, with 31.4 by 9 % more. The wing's yawing moment with sideslip at zero lift, −0.0?15, is Ferguson's −0.0315 (the model's wing strips give their own). *(ii)* The XV-15 rotor's hover test [3] gives its power as the report's curve fit through the wind-corrected data, C_P/σ against C_T/σ ($\sigma = 0.0891$), and its collective from the actuator's position, whose reference to a blade station is not given; it is compared here as a gradient. *(iii)* The XV-15's level-flight power [4] is published with neither the gross weight nor the altitudes; the flight envelope puts most of the programme between 2000 and 10 000 ft density altitude [5], so the model is flown at 13 000 lb at sea level and at 10 000 ft, which bracket it. Nor is the landing gear's position given; the gear is up through-

Table 1: Sources and what they contribute.

Source	Content used
NASA CR-114614 [1]	Bell's Model 301 (XV-15) simulation data: wing–pylon, fuselage and empennage tables, downwash, control mixing and rigging, governor, mass and inertia
NASA CR-166536 [26]	Ferguson's generic tilt-rotor simulation: Bell's XV-15 data revised against its flight tests (mass and inertias, tail, ailerons)
NASA TM-89428 [27]	XV-15 bare-airframe frequency responses identified from flight tests, hover and cruise
NASA TM X-62407 [2]	XV-15 familiarisation: controls, flaperons, conversion range
NASA TM-86833 [3]	full-scale XV-15 rotor in hover: power against thrust, collective, blade twist, chord and sections
NASA TM-84222 [4]	XV-15 level-flight rotor shaft power from flight test by nacelle angle and flap (fig. 20)
12th ERF, paper 14 [6]	V-22 design data, rotor geometry and sections, wing, download against flap (XV-15 flight and V-22 0.658-scale test), airframe polar, hover ceiling, flight envelope
16th ERF, II.7.3 [7]	V-22 control system, conversion corridor, rotor speeds, airframe lift share and pitch attitude through the conversion, power required
NASA/TM-2016-219070 [8]	JVX (0.656-scale V-22) rotor characteristics, precone
Johnson, AHS 2001 [9]	1/4-scale V-22 rotor (TRAM): sections, δ_3, chord and twist
V-22 fact sheet [10]; GlobalSecurity [11]	weights, fuel capacity, crew; the MV-22's tanks and their transfer sequence
NACA R-824 [12]	64-series sections' lift, stall and drag (the sections' parameters)
NASA TN D-234, NACA TR 1319 [13, 14]	ground effect in forward flight; ring-vortex wake
Peters and He [15], via [16]	finite-state dynamic wake
NASA/TP-2005-213477 [30]	Johnson's vortex ring state model of a rotor's mean inflow in descent and its time lag; the flight tests' VRS encounters, a tiltrotor rotor's inflow in descent [31], his V-22 encounter

out (Section 5.2). *(iv)* Dunford's conversion charts [7] give no weight, altitude or flap setting. Their airplane-mode pitch attitude, 13.6° at 108 kt falling to 0.2° at 170 kt, asks for a lift coefficient of 0.82 at zero angle of attack with Boeing's lift slope [6]: about 31 500 lb at sea level, below the MV-22B's empty weight. The comparisons run at 34 000 lb, sea level. *(v)* Boeing's hover ceiling and flight envelope [6] are for the 1986 design with 6150 shp engines and a standard day.

3 The model

3.1 Implementation

A Python script writes the model from the transcribed tables (Table 2), the XV-15 or the V-22 from the same code with each aircraft's data. JSBSim evaluates aerodynamic functions on demand, so the function graph must be free of cycles; every quantity that depends on its own past, the gimbals' tilt and rate, the rotor azimuth and speed, the wake states, the nacelle angle, the engines' states, the governor's and the augmentation's integrators, therefore lives in flight-control-system channels, which JSBSim runs once per frame in a fixed order and which can write properties back. The blade elements, the gimbals and the drive train integrate explicitly at 480 Hz, the rate FlightGear runs the model at: at 104 % rotor speed 5.2° of azimuth per step. The rest runs every few frames, which JSBSim's channels allow: the wake states with their gains and forcing (time constants of a tenth of a second and more) and the controls, mixing, servos, governor and augmentation, every fourth (120 Hz), the airframe with its wake and ground-effect tables, the engines and the outputs to FlightGear every eighth. In FlightGear every property read takes a lock, so a step costs what it reads; with these rates and the blade elements' terms arranged for it (the first wake shape's and the flap angle's share of the normal velocity taken per blade, the angle of attack into tables carried on to ±360° instead of wrapped) a step reads about 3200 properties, 4300 at the FDM rate throughout; the ceilings, speeds and download are the same to five digits, the trims within 0.2 % of power and 0.04° of attitude. The same XML runs in FlightGear and in the offline harness used here; FlightGear sets only the pilot's controls and reads the outputs.

Table 2: The generated V-22 model: JSBSim system files, their content and the number of named aerodynamic functions or control-system components.

File	Content	n
`-aero.xml`	the aerodynamic functions: blade segments of both rotors, gimbal moments, wing strips in the wakes, fuselage, empennage; the six axes	622
`-rotors.xml`	gimbals, rotor azimuth, Peters–He wake states of each rotor with their gains and forcing, the vortex ring state	123
`-airframe.xml`	ring-vortex wake at the wing, stabiliser and fins; ground effect	164
`-drive.xml`	engines, torque limit, interconnected drive and rotor speed, rotor brake, fuel	37
`-controls.xml`	pilot controls, mixing, nacelles and conversion protection, flaperons, governor, augmentation, hover hold, autopilot, servos	300
`-outputs.xml`	properties for the 3-D model and the instruments; the crew alerts' conditions	120

3.2 Proprotors

Each rotor's three blades are rigid in bending and sit with their precone (2.5° [8, 3]) on a gimbal, so the disc tilts as a whole, as on both aircraft. The gimbal's two degrees of freedom, in the non-rotating frame, are integrated from the blades' aerodynamic flapping moments about the hub, the gyroscopic coupling of the shaft's rotation (taken exactly, as a rotation of the tilt each step), the airframe's angular rates and a hub spring; each blade's pitch takes $\tan\delta_3$ of its own flapping, $\delta_3 = -15°$ [1, 9]. The blades' inertia about the gimbal is the XV-15's Lock number's (3.8), scaled. Eight segments of equal annulus area from $0.15\,R$ (V-22; $0.1\,R$ XV-15) see the air velocity at their mid-point from the full kinematics: the aircraft's motion, the nacelle's tilting rate, the rotor's ro-

tation, the gimbal's tilt and rate and the induced velocity. The left rotor is the right one's mirror image and is computed as the right rotor of the mirror-image aircraft.

Each blade twists in its first torsion mode, a state of its own. The mode is the XV-15 metal blade's, from its mass, pitch inertia and torsional stiffness at 51 stations and its control system's stiffness, NASA's model of the blade [23]: from the pitch bearing out, on the pitch link's stiffness through its horn, its frequency is 3.3/rev at 589 rpm, and the rigid blade's pitch on the control system 5.3/rev at 458 rpm, Johnson's 5.2 [24]. It is driven by the sections' moments about the elastic axis (each section's moment at its quarter chord, $-0.8\,c_{l_i}/4$ with the zero-lift angle's factor, and its normal force at the quarter chord, whose line is swept 1° aft, crossing the axis at $0.75\,R$), by the propeller moment of the blade's whole pitch, $-\Omega^2 I_\theta \sin 2\theta/2$ along the span, which turns a twisted proprotor blade's sections toward flat pitch, and by its centre of gravity's offset under the gimbal's flapping; it is damped by the structure and, mostly, by the sections' pitching (13 % of critical). The XV-15's tips twist 4° nose down in the hover and 5.6° in cruise at 180 KCAS. The V-22's composite blade is not published, but the blade of its dynamically scaled 1/5-scale model was ground vibration tested: its first torsion, scaled to the full size, is 50.7 Hz [25]. The V-22's mode is the XV-15's blade stretched to the V-22's size and blade inertia, its torsional and control system stiffness raised together to that frequency, 7.7/rev; its tips twist about 1°. The blades' bending is left rigid: their first flap and lag modes are near 1.4/rev, and under the trims' loads they would cone 0.1° beyond the precone in the hover, which balances its thrust, and 1.0° short of it in cruise, and lag 0.2° (blademodes.py).

Neither rotor's section tables are public: the XV-15's metal blades use NACA 64-series sections [3], the V-22's Boeing's XN sections of 28 % thickness at $0.2\,R$ to 9 % at the tip [6, 9]. Each segment's section is built as C-81's equations build the NACA 0012, with the parameters of a cambered section of its thickness and design lift [12]: a lift slope and zero-lift angle from the camber, a maximum lift falling with thickness and Mach number, a trailing-edge stall that rounds over into the maximum and falls gently past it to a flat plate's normal force, Korn's drag divergence. The stall's shape, its maximum lift ($C_{l\max} = 1.55$ at zero camber), the separated drag and the minimum drag (0.0065 at 12 % thickness) are *fitted* to the XV-15 rotor's hover power with the blades twisting (Section 5.1). The V-22's twist, 47.5° from the root to the tip, and its chord, 34.3 in extrapolated to the centre to 22 in at the tip, are Boeing's [6, 9]; the tip loss factor is 0.97.

3.3 Inflow and wake

Each rotor has its own Peters–He finite-state wake [15, 16], three axisymmetric states and the first harmonic pair, in the shaft's frame as the nacelle tilts. Ground effect scales the induced velocity by $(1+0.3(R/Z)^2)^{-1/2}$ in the hover, with Z the hub's height, carried into forward flight with Heyson's interference ratios [13]. Each rotor's downwash at the wing, the horizontal stabiliser and the two fins comes from a skewed cylinder of ring vortices leaving the disc [14], contracting below it in the hover: averaged over four spanwise strips of each wing half, the stabiliser and each fin, and tabulated against the wake's skew, its direction in the disc's plane and the nacelle angle, which moves the hub and turns the disc.

Momentum theory, on which the wake's uniform state rests, has no vortex ring state. In descent each rotor's uniform state follows Johnson's model of the mean inflow instead [30], an extension of momentum theory built on the wind tunnel and flight tests of rotors in descent: a cubic removes momentum theory's singularity at ideal autorotation, and three more add the vortex ring's increment, so that the total flow through the disc, $V_z + v$, rises as the descent rate rises between his points N and X, the boundary of the instability the flight tests met. Where it departs from momentum theory, descending faster than $0.2\,v_h$ at less than $0.95\,v_h$ edgewise, the uniform state's mass flow parameter is set for his steady inflow, $v_h/v(V_z/v_h, V_x/v_h)$, in place of momentum theory's $\sqrt{V_x^2 + (V_z + v)^2}$, with V_z and V_x the rotor's own axial and edgewise velocities: each rotor enters the vortex ring on its own. Its scale v_h is the wake's own: the steady uniform inflow times its mass flow, from the uniform state's row of the wake's L matrix and the forcing. The model's momentum theory is then Glauert's in V_z/v_h and V_x/v_h exactly, and Johnson's joins it at the region's edges. While his total flow is down through the disc the mass flow keeps momentum theory's form on the present inflow, offset to his steady state, so that the wake keeps its own dynamics and meets momentum theory at the edges exactly (a step of collective in vertical descent moves the uniform inflow 63 % of the way in 0.117 s at $0.19\,v_h$, outside, and 0.117 s at $0.21\,v_h$, inside); past ideal autorotation, where that form has no stable state, it is his own, the two meeting there with the same slope; in the hover this v_h is 1.065 times the thrust's ideal $\sqrt{T/2\rho A}$ (the tip loss and the nonuniform inflow: Johnson's induced power factor κ, which he takes as 1.05 to 1.2 against the tests). His curve is tabulated against V_z/v_h and the coordinate his algorithm moves N in, $(1 - (V_x/0.95\,v_h)^2)^{0.2}$, 5300 points within $0.022\,v_h$ of his algorithm. The vortex ring's increment follows with his first-order lag, $0.7/\lambda_h$ revolutions (1.2 s for the V-22 in the hover), which he fitted to a helicopter's flight test. Johnson gives no model of the vortex ring's unsteadiness, whose measured frequencies and amplitudes show no trend [30]; none is added.

3.4 Airframe

The wing–pylon, fuselage, horizontal stabiliser and fin data are Bell's tables for the XV-15 [1]: the wing's lift and drag against angle of attack, flap setting and mast angle, the fuselage's forces and moments in equation form, the stabiliser's and fins' coefficients with their blockage. The tail and the ailerons are as Ferguson revised them against the XV-15's flight tests for NASA's generic tilt-rotor simulation [26], which found Bell's elevator, rudder and aileron effectiveness and the tails' dynamic pressure questionable: the wing's downwash at the tail by mast angle and flap and its lag behind the wing, the rotors' wake over the tail (a share of their induced velocity by angle of attack, speed and mast angle, for the ring-vortex wake's there; the V-22 reads it at the airspeed that gives the XV-15 the same wake skew, its own times the ratio of the two rotors' induced velocities in the hover, 0.80), the stabiliser's dynamic pressure at 0.8 of the free stream's, the elevator's and the rudders' effectiveness (0.518, 0.27) with their Mach number factor, the stabiliser's lift slope against Mach number, and the ailerons' rolling moment (0.006 per degree), yawing moment and loss of lift; the XV-15's fins are

his (20.25 ft^2 each), and so is the drag the XV-15 build adds to Bell's coefficients from the flight tests: the spinners' and the pylons' interference with the wing, by mast angle (the V-22's fitted drag, below, stands for its own). Bell's wing tables are against the fuselage's angle of attack, the XV-15's 3° of wing incidence in them [1]; the V-22's wing is set as the XV-15's. Bell gives no 20° flap: the model interpolates between his flaps-up and 40° tables. Ferguson's wing tables, which have one, are not used. From about −8° to the stall they lift 0.12 to 0.18 more than Bell's flaps up and about 0.2 more with flap, with that lift's induced drag, and his own table of the lift slope at zero lift, defined as the difference of two of these tables' values, is Bell's tables' exactly (0.0799 and 0.057 per degree flaps up, airplane and helicopter mode), not his (0.105 and 0.09): the lift was raised afterwards, in the tuning of his simulation against the flight tests [26], whose rotors act on the wing otherwise than the wake here. Tried in the model, his tables brought the conversion's power and elevator nearer the flight test but airplane mode, where the rotors' wake hardly reaches the wing, further from it (about 2° of elevator and 8 points of power with 20° of flap), and gave the V-22 a lift coefficient of 0.34 at zero angle of attack for Boeing's 0.18 (Section 5.5). For the V-22 the wing's are applied to its own area, span and chord (382 ft^2, 45.8 ft [6]), the empennage's to its surfaces, and the fuselage's scaled by the length ratio squared and cubed. Each wing half is four strips in the rotors' wakes, with the local flow's angle of attack. In the hover the wakes push the wing down; the download is Bell's flap-dependent coefficient in the wake's velocity times a factor *fitted* to the XV-15's download in flight for each flap setting (Section 5.4). Bell's coefficients miss part of the V-22's parasite drag: 11.5 ft^2 is added, *fitted* to Boeing's airframe polar, flaps and gear up (Section 5.5). The landing gear adds Bell's drag areas, scaled, when extended.

3.5 Flight controls

The cockpit controls' mixing is Bell's for the XV-15 [1], in its control travel: the thrust control lever sets both rotors' collective (its slope falling to zero in airplane mode), the lateral stick their differential collective, the longitudinal stick their longitudinal cyclic, the pedals their differential cyclic, each phased with the mast angle, while the elevator, flaperons and rudders follow the stick and pedals in all modes. The nacelles move at up to 8°/s from 97.5° (7.5° aft of vertical) to 0, their command held within the conversion corridor against equivalent airspeed [7]. The flaperons' schedule is not public; they are at 67° in the hover for the least download [6] and at 40° through the conversion with the nacelles up (the corridor's lower boundary is the wing's stall at 40° of flap [7]), out by 160 kt. Elsewhere the V-22's come from the flight state: the least deflection that makes the wing's lift, from Bell's wing data [1], with a margin of 1.3 g to its stall and at no more than AUTO NAC's pitch in level flight (Section 5.10), at most 40° (Bell's 67° setting has the 40° one's maximum lift), in ENG OUT's glide too, where at 60 500 lb it costs 14 % of the reach and its flares land better. At any lift the least flap has the least drag in Bell's data, but by the wing's drag alone the pitch rose 1.5–2.5° at each nacelle angle, AUTO NAC kept the nacelles up longer, and its schedule took up to 1770 hp more; with the pitch limit the nacelles come down sooner and level flight at 150–160 kt along it takes about 2000 hp less than the earlier estimate's. The XV-15's build keeps that estimate: 40° in airplane mode below 120 kt, retracting by 200 kt. A collective governor holds the rotor speed's reference, 104 % in helicopter and conversion modes and 84 % in airplane mode [7], proportional and integral within Bell's collective limits.

The V-22 flies by wire and holds attitude in all modes [7]; its control laws are not public. The FlightGear aircraft has a stability augmentation of a common rotorcraft structure: rate damping in three axes for a 0.5 s rate time constant in the hover, its gains falling with the dynamic pressure as each axis's surface gains power (each axis's schedule by its own surface's); attitude command and heading hold below 50 kt with the nacelles above 60° and rate command with attitude hold above (switching at 55 kt accelerating and 45 kt slowing, so that a speed held near 50 does not switch back and forth), switched without a transient because both share one reference [22]; in yaw below 50 kt rate command, direction hold, the pedals commanding a yaw rate (30°/s at full travel) held against the wind by an integrator, the hover's response type in ADS-33E [17]; a feed-forward of the level-flight trim's stick against nacelle angle, forward airspeed and the flaperons, from the model's own trims, so that the stick's centre is trimmed flight through the conversion at whatever the flaperons are; and integrators that take up the rest. A hover hold (Section 5.9) and an autopilot (Section 5.10) work through it. The comparisons of Section 5 fly the bare airframe, the augmentation off.

3.6 Engines, drive and mass

The engines are not modelled in detail: the lever sets each one's power demand from flight idle to its rating (6150 shp each [6]), falling with altitude as $\delta/\sqrt{\theta}$; the engine follows with a 0.6 s lag, and a topping governor takes power off above 2 % of overspeed. The interconnected drive gears both rotors to one speed, with the rotors', the power turbines' and the interconnect shaft's inertia; the drive's torque limit, shared between the running engines, is *fitted* to Boeing's sea-level hover ceiling (9100 hp at 100 % rotor speed; Section 5.6). A start brings each engine from rest to flight idle in 30 s. The MV-22B's five tanks (1448 US gal [10, 11]) are drained in the transfer sequence's order. Each nacelle, 7000 lb with its engine, gearbox and rotor (15 % of the gross weight, as the XV-15's pylons [1]), tilts with its centre of gravity and inertia. The reference loading is 47 500 lb, the 1986 design's maximum vertical take-off weight [6], at a centre of gravity under the hubs in helicopter mode: the empty weight (33 140 lb [10]), three crew, the tanks full but for the right aft sponson and the rest as cargo. The V-22's inertias are the XV-15's of Bell's data [1] scaled by mass and dimensions. The XV-15 build's mass, centre of gravity and inertias are Ferguson's for the aircraft as it flew, with its modifications and instrumentation [26] (Bell's pitch and yaw inertias are a third lower; Tischler traced the simulation's too sensitive pitch and yaw responses to them [27]). Each nacelle is two point masses on its shaft, their spacing carrying its own inertia across the shaft, so that the aircraft's inertia changes with the tilt: the XV-15's as Ferguson's does, within 0.5 %. Table 3 lists the principal data.

Table 3: Principal data of the two builds.

Item	V-22	XV-15
Rotor radius, ft	19.04	12.50
Blades per rotor	3	3
Precone, deg	2.50	2.50
δ_3, deg	-15	-15
Chord root (extrapolated) / tip, in	34.3 / 22.0	14.0 / 14.0
Rotor speed 100 %, rpm	397	589
Rotor speed, helicopter/airplane, %	104 / 84	100 / 88
Wing area, ft^2; span, ft	382; 45.83	181; 32.17
Engines, shp each	6150	1550
Drive torque limit at 100 %, hp	9100 (fitted)	2500
Nacelle angle range, deg; rate, deg/s	0–97.5; 8	0–95.0; 8
Reference gross weight, lb	47500	13000
Empty weight, lb	33140	–
Fuel, US gal (tanks)	1448 (5)	–

Table 4: The isolated XV-15 rotor in hover: power and figure of merit at the model's thrust.

θ_{75}	C_T/σ	C_P/σ, model / test		Δ %	FM, model / test	
10.0	0.075	0.00621	0.00601	+3.4	0.704	0.728
11.0	0.086	0.00719	0.00708	+1.5	0.738	0.750
12.0	0.096	0.00828	0.00825	+0.3	0.763	0.766
13.0	0.107	0.00949	0.00948	+0.1	0.779	0.780
14.0	0.118	0.01081	0.01087	−0.6	0.789	0.785
15.0	0.128	0.01224	0.01243	−1.5	0.792	0.780
16.0	0.138	0.01378	0.01396	−1.3	0.789	0.779
17.0	0.148	0.01543	0.01575	−2.0	0.781	0.765
18.0	0.157	0.01718	0.01769	−2.8	0.767	0.746
19.0	0.166	0.01908	0.01945	−1.9	0.746	0.731
20.0	0.173	0.02113	0.02129	−0.8	0.717	0.712
21.0	0.179	0.02335	0.02322	+0.6	0.683	0.687

4 Method

The evaluation drives JSBSim from Python with the same files as FlightGear, at 480 Hz, the augmentation off and the gear up (in the hover its drag is nil). A trim solves for the collective, the cyclic or elevator, the pedals, the angle of attack and the roll that zero the six body accelerations, averaged over whole rotor revolutions (Hann-weighted), with the airframe held and the rotor at the governed speed, by Newton's method with a backtracking line search. The programme:

- *the XV-15 rotor in hover*: the XV-15 build held far from the ground at the test's tip Mach number (589 rpm, 1000 ft), swept in collective; the right rotor's thrust and power;
- *XV-15 level-flight power*: trims at 13 000 lb at sea level and 10 000 ft by calibrated airspeed for each of the test's nacelle and flap settings, the flaperons at 0.625 of the flap [2];
- *download*: hover trims out of ground effect at sea level, the XV-15 at its flight tests' flap settings, the V-22 at 67°;
- *XV-15 frequency responses*: the XV-15 build trimmed at the flight tests' conditions, hover and cruise at 180 KCAS and 10 000 ft, swept in one control at a time from 0.08 to 20 rad/s over 240 s at the trim's power, with a pilot's attitude hold (in cruise also holding the speed by the attitude); the response from the surface's deflection, as in the tests, to the rates, sideslip and normal acceleration, against the transfer functions identified from the flight tests [27] by Tischler's cost J over their fitting ranges (1 dB of magnitude weighs as 7.6° of phase; under about 100 is a good match);
- *V-22 airframe polar*: the airframe's own forces in airplane mode at 198 KTAS (Mach 0.3) and 3000 ft, flaps and gear up, the rotors at zero thrust;
- *V-22 performance*: the maximum level speed in airplane mode at 84 % rotor speed, where the power required meets the engines' power or the drive's limit; the hover ceiling out of ground effect at 100 % rotor speed (the 1986 chart's); trims through the conversion at 34 000 lb by nacelle angle;
- *handling* with the augmentation on: single keyboard steps in the hover, a hands-off conversion, an engine start;
- *the hover hold*: its stops from up to 39 kt, ADS-33E's position hold turn in up to 35 kt of wind, its stick and lever;
- *the autopilot*: coupled from trims at 1000 ft, an altitude change, a turn, the conversion down and up under AUTO NAC; approaches to a hover and landings, a route's, go-arounds; a route's turns; terrain following over profiles of terrain set under the aircraft as it flies, with the plan of the FlightGear script mirrored (every 0.25 s, and with the timer up to a frame late);
- *the vortex ring state*: both builds trimmed in steady descents in helicopter mode at 1000 ft, out of ground effect, at 0 to 60 kt; Johnson's V-22 encounter, from his descent with his collective, the augmentation on and the rotor speed held.

5 Results

5.1 The XV-15 rotor in hover

From $C_T/\sigma = 0.08$ to 0.18 the rotor needs −2.8 to +3.4 % of the test's power at the same thrust (Fig. 1, Table 4), the figure of merit peaking at 0.79 at $C_T/\sigma = 0.13$ against the test's 0.79 at 0.12, and the rotor reaches $C_T/\sigma = 0.184$ at 23° of collective, where the test, which stopped at 0.181, did not go. The sections' stall and minimum drag were fitted here, so this checks the shape of the curve rather than predicts it. Below $C_T/\sigma = 0.075$ the model needs 5–19 % more power than the test, more the lower the thrust: the profile power of the inboard sections, whose thick XN-like stand-ins have more drag at low lift than the 64-series. The thrust's gradient against collective is 0.0102 per degree against the test's 0.0085 (20 % steeper): the test's collective is the actuator's, and the blades twist under load, nose down the more the higher the collective as the propeller moment grows; their torsion mode takes 14 points of the 34 that rigid blades left. Past the maximum thrust the blades stall progressively and the gimbal whirls by a few degrees; the model's power there (the points that turn back in Fig. 1a) has no test to compare with.

5.2 XV-15 level-flight power

The XV-15 build carries the drag Ferguson added from the flight tests to Bell's coefficients, which come from wind tunnel tests of the clean model [26]: the spinners' (1 ft^2 each along the mast, 5.5 times the square of the sine of their angle of attack across it) and the pylons' interference with the wing, 13.5 ft^2 with the nacelles up, falling to 1 ft^2 in airplane mode (Section 3). In airplane mode the model then needs less power than the test at both altitudes still: −22 to −16 % with 20° of flap at sea level and, flaps up, −33 to −23 % at

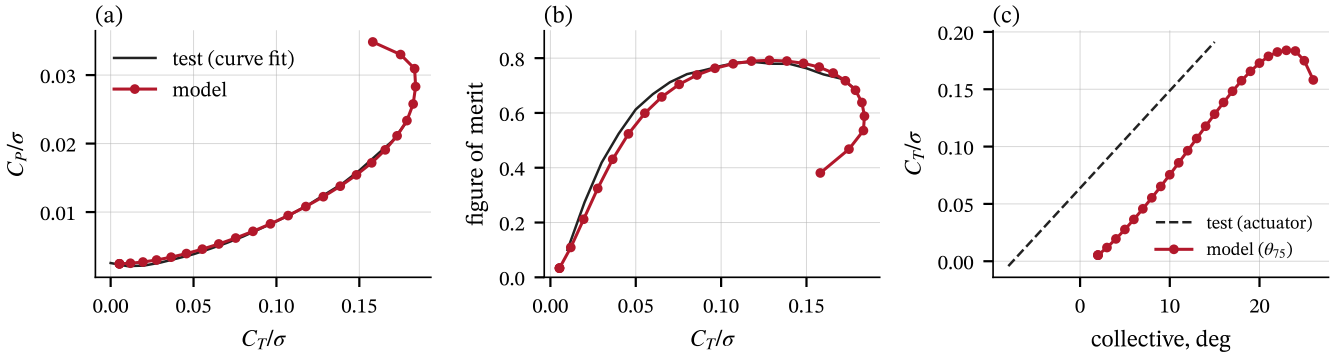


Figure 1: The isolated XV-15 rotor in hover at a tip Mach number of 0.69 against NASA's full-scale test (TM-86833 figs. 13, 14, 17): (a) power against thrust, (b) figure of merit, (c) thrust against collective (the test's from the actuator's position, the model's commanded at 0.75 R, before the blades' twist).

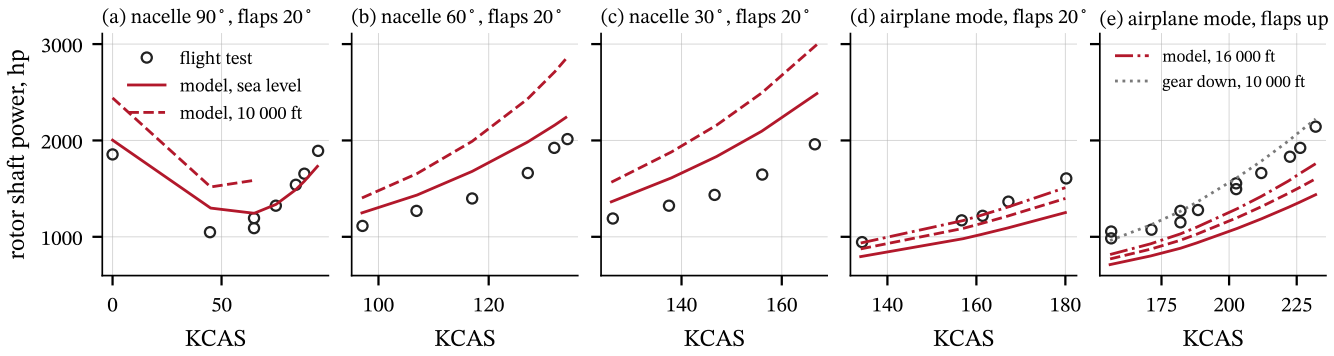


Figure 2: XV-15 rotor shaft power in level flight against the flight test (TM-84222 fig. 20) at 13 000 lb: the model at sea level and at 10 000 ft, which bracket the test's unpublished altitudes, and in airplane mode (d, e) at 16 000 ft, where the set's fastest point was flown; gear up, in (e) also down.

Table 5: XV-15 level-flight power, model/test −1 at the test's points.

Nacelle, flaps	KCAS	sea level, %	10 000 ft, %	16 000 ft, %
90°, flaps 20°	0–94	−8 to +24	+32 to +45	
60°, flaps 20°	97–134	+11 to +20	+26 to +46	
30°, flaps 20°	126–167	+15 to +28	+32 to +52	
0°, flaps 20°	134–180	−22 to −16	−13 to −6	−6 to +1
0°, flaps up	156–232	−33 to −23	−27 to −16	−22 to −10
0°, flaps up, gear down	156–232	−16 to −1	−8 to +10	

Table 6: XV-15 trim elevator, model less flight test at the test's points: the mean (and range), deg trailing edge down.

Nacelle, flaps	KCAS	sea level, deg	10 000 ft, deg	16 000 ft, d
90°, flaps 20°	44–97	−2.3 (−4.6 to −0.4)	+0.4 (−0.2 to +1.4)	
60°, flaps 20°	97–135	−3.2 (−4.6 to −2.0)	−0.8 (−2.4 to +0.1)	
30°, flaps 20°	128–169	−3.2 (−3.4 to −3.0)	−2.4 (−2.7 to −2.1)	
0°, flaps 20°	139–183	−0.5 (−1.4 to +0.5)	−0.3 (−1.2 to +0.6)	−0.2 (−1.2 to +0.
0°, flaps up	160–179	−0.1 (−0.6 to +0.6)	+0.0 (−0.4 to +0.7)	+0.2 (−0.3 to +0.

sea level and −27 to −16 % at 10 000 ft (without them −46 to −35 %: Fig. 2, Table 5). The figure's fastest flaps-up point was flown at 16 000 ft (300 KTAS), and there the model needs −22 to −10 % flaps up and −6 to +1 % with 20° of flap: the unpublished altitudes account for some of the shortfall, with the flaps up not for all of it. With the gear extended, its 4 ft^2 of drag [1] brings the flaps-up power to −8 to +10 % at 10 000 ft; the test's gear position is not given, but in airplane mode it would have been up. The V-22's coefficients needed 11.5 ft^2 more to meet Boeing's polar for the same reason (Section 5.5). Through the conversion, nacelles at 60° and 30° with 20° of flap, the model now needs more power than the test at both altitudes, +11 to +20 % and +15 to +28 % at sea level, +26 to +46 % and +32 to +52 % at 10 000 ft. Without Ferguson's drag the test lay between the two altitudes there, but only because the drag's absence hid power the model makes in excess: Ferguson needed the drag with Bell's rotor, which under-predicted the power in forward flight [26], and the blade elements here do not. The excess is in the conversion alone, with the rotors' wakes over the wing: the trim (below) points there too.

In helicopter mode the model's power falls too fast with speed: +24 % at 45 kt, −8 % at 94 kt at sea level, +8 % in the hover. Bell's own simulation used an empirical profile power for the rotor in this regime; the blade elements here give a translational lift that lowers the induced power faster than the test, and the wing in the slipstream at a high angle of attack, with 20° of flap, sees the wake model's velocities, whose sweep off the wing through the transition is not checked by any test. At 94 kt Ferguson's spinner and pylon drag take two thirds of the shortfall away.

The same report gives the trim elevator of those flights (Fig. 3, Table 6), which checks the pitching moments' balance and the flow at the tail. In airplane mode the model trims within half a degree of the flight test, −0.1 and +0.0° on average flaps up, −0.5 and −0.3° with 20° of flap (+0.2 and −0.2° at 16 000 ft). Through the conversion it needs less elevator than the aircraft did, at sea level by more than at 10 000 ft: −3.2 and −0.8° at 60° of nacelle, −3.2 and −2.4° at 30°, −2.3 and +0.4° in helicopter mode. Bell's simulation was within two or three degrees there; NASA traced its errors to the

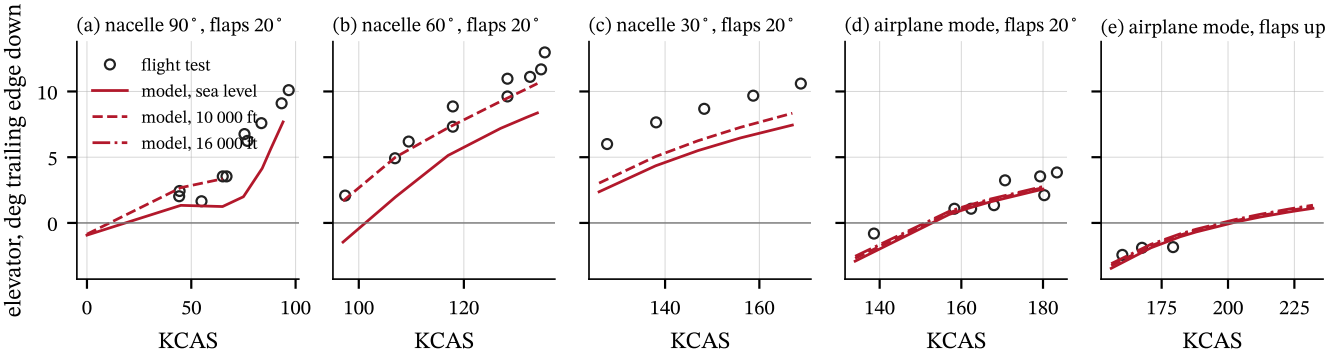


Figure 3: XV-15 trim elevator in level flight against the flight test (TM-84222 fig. 21) at 13 000 lb, the model at sea level and at 10 000 ft, in airplane mode (d, e) also at 16 000 ft.

Table 7: The XV-15's frequency responses against the flight tests': Tischler's cost J over each fitting range (Bell's simulation's in cruise), the mean magnitude error and the mean phase error.

cond.	response	rad/s	J	Bell	ΔdB	$\lvert\Delta\phi\rvert$
hover	p/δ_a	0.2–9	2238	–	−2.0	49
hover	r/δ_a	0.1–7	4787	–	+5.4	79
hover	r/δ_r	0.1–3	1019	–	+0.8	29
hover	q/δ_e	0.2–5	142	–	+1.5	6
hover	a_z/δ_c	0.1–3	676	–	−4.3	17
cruise	p/δ_a	0.1–8.5	113	60	+0.1	16
cruise	β/δ_r	0.1–3.5	387	131	−4.1	6
cruise	q/δ_e	0.2–7	503	3768	−0.5	20
cruise	a_z/δ_e	0.3–10	462	187	−0.8	22

downwash at the tail, which Ferguson then revised [4, 26]. The tail here has his downwash and his rotor wake over it: between the two rotors the wake is an upwash, which the ring-vortex wake had made several times as strong at 60° (1.8° at 118 kt against his 0.3°). It brought the conversion's trim 2° toward the flight test, and his pylon drag, high on the wing, about 3° more. What remains, a nose-up moment the model lacks only with the nacelles up, points, as the conversion's excess power does, to the rotors' wakes over the wing rather than to the tail.

5.3 XV-15 frequency responses

Tischler identified the XV-15's responses to its control surfaces from frequency sweeps flown with the augmentation on, whose part the surface's deflection takes out of the loop [27]; the model is flown the same way (Section 4). With Ferguson's revised data the cruise's lateral responses follow the flight test (Fig. 4, Table 7): the roll response to the ailerons within its scatter (J 113), and the sideslip's to the rudders near the flight's dutch roll frequency, its magnitude about 4 dB low (J 387). With Bell's 1973 tail and inertias, which the model first had, M_α was $-18.9\,\mathrm{s}^{-2}$ and M_{δ_e} 21 deg/s^2 per degree.

The pitch response in cruise is still the poorest (J 503, normal acceleration 462): its short period is near 3 rad/s. The derivatives tell why: M_α $-8.7\,\mathrm{s}^{-2}$ against the flight's -2.95, M_{δ_e} 10.4 against 7.38 deg/s^2 per degree, Z_w -0.86 against $-0.89\,\mathrm{s}^{-1}$. The stabiliser gives the force Ferguson's data give, and at Bell's own condition, 260 kt at sea level with an aft centre of gravity, the model's static stability is Bell's (C_{M_α} -1.32 against -1.32). At the flight test's condition its parts are the tail's -2.71 per radian, the wing's and the fuselage's +0.79 and the rotors' +0.21. Tischler found that Bell's simulation had the flight's static gain, its error being the pitch inertia it lacked [27]; with Ferguson's inertia, the model's elevator is 1.4 and its stiffness 2.9 times the flight's. A larger pitch inertia would bring the elevator's power to the flight's and leave the stiffness twice the flight's; a weaker stabiliser would bring both. The flight tests' responses alone cannot tell which, and the model keeps the published data.

Table 8: Download in the hover, % of the rotors' thrust.

Aircraft	flap, deg	model, %	test, %	Δ
XV-15	0	16.6	16.4	+0.2
XV-15	40	13.3	13.0	+0.3
XV-15	60	11.6	10.4	+1.2
XV-15	75	10.4	10.5	−0.1
V-22	67	11.8	9.2	+2.6

In the hover the pitch response follows the flight test (J 142) and the yaw response to the pedals nearly (J 1019). The roll response to the ailerons, which in the hover move the rotors' differential collective with them, has the flight's magnitude, but below 0.5 rad/s, where the flight test found an unstable lateral oscillation and a zero in the right half plane, its phase differs (J 2238); the yaw that comes with it is twice the flight's (J 4787). The heave response to the power lever is 4 dB low, with a notch near 1.3 rad/s where the governor takes back the lever's collective while the engines' power follows the lever (J 676).

5.4 Download

The download factor of each of Bell's flap settings was fitted to the XV-15's flight-measured download, which the model then follows within 1.2 point of thrust (Fig. 5a, Table 8). Carried to the V-22 with its own wing and rotors, at the hover's 67° of flaperon, it gives 11.8 % against the 0.658-scale test's 9.2 %: the V-22's flaperons, which fold down over a larger part of the chord than the XV-15's flaps, take more of the wing out of the wake than the XV-15's tables know.

5.5 V-22 airframe polar

Bell's wing and empennage tables applied to the V-22's surfaces give it a lift coefficient of 0.22 at zero angle of attack and a slope of 0.096 per degree, against Boeing's 0.18 and about 0.093 (Fig. 5b, Table 9). Its drag, with the 11.5 ft^2 fit-

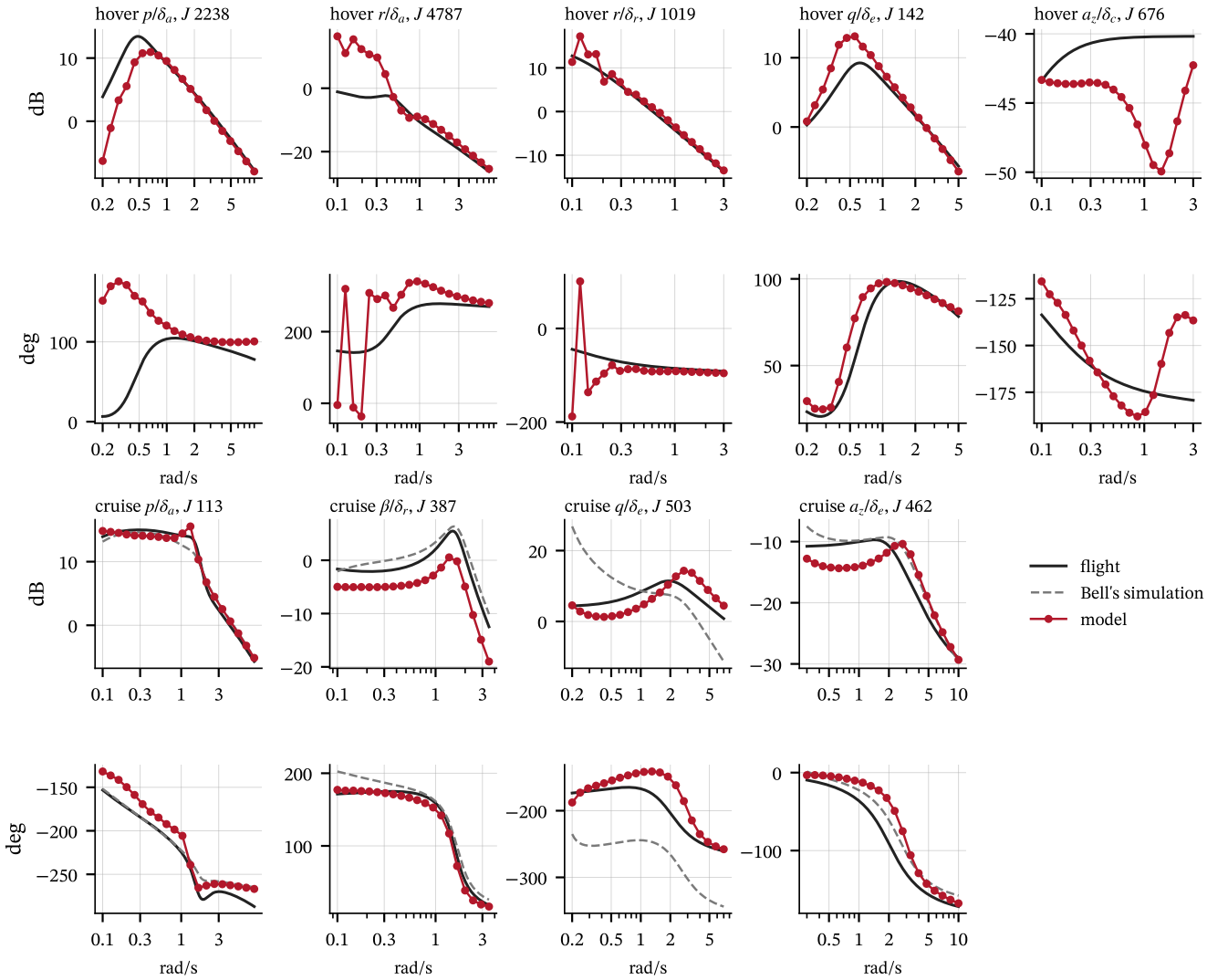


Figure 4: The XV-15's bare-airframe frequency responses over the flight tests' fitting ranges: the transfer functions identified from the flight tests (TM-89428 tables 4.1 and 5.1), Bell's simulation's in cruise, and the model's. Hover out of ground effect at 13 000 lb; cruise at 180 KCAS and 10 000 ft. Rates in deg/s, sideslip in deg and normal acceleration in g per degree of the surface or per cent of the power lever; NASA's signs. J: Tischler's cost.

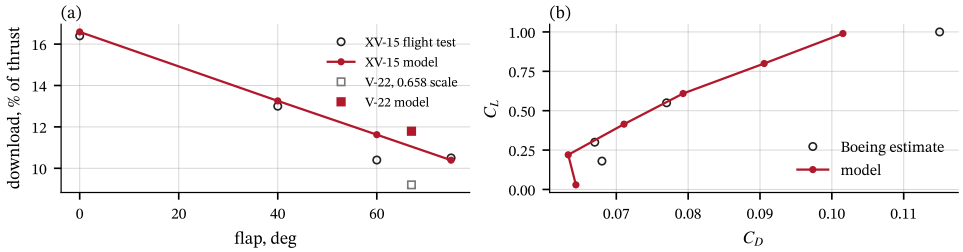


Figure 5: (a) The wing's download in the hover against flap angle: the XV-15 (13 300 lb, flight test [6] fig. 18) and the V-22 at 67° (0.658-scale rotor and wing test). (b) The V-22 airframe's lift and drag in airplane mode, flaps and gear up, against Boeing's full-scale estimate [6] fig. 28.

Table 9: V-22 airframe drag coefficient at Boeing's lift coefficients, on 382 ft^2; Δ in drag counts.

C_L	C_D model	C_D test	Δ, counts
0.18	0.0635	0.0680	−45
0.30	0.0665	0.0670	−5
0.55	0.0768	0.0770	−2
1.00	–	0.1150	–

ted here, is 0.0634 at $C_L = 0.2$ against 0.068; the model's polar is flatter than Boeing's estimate, whose drag rises faster beyond $C_L = 0.6$.

5.6 V-22 performance

The drive's torque limit was fitted to the hover ceiling at sea level, 47 700 lb, where Boeing's chart has its knee: the transmission limits there, the engines above. With it the ceiling is 42 230.0 lb at 8400 ft (−2 %) and 32 600.0 lb at 16 000 ft (+5 %), where the engines' lapse sets it (Fig. 6, Table 10).

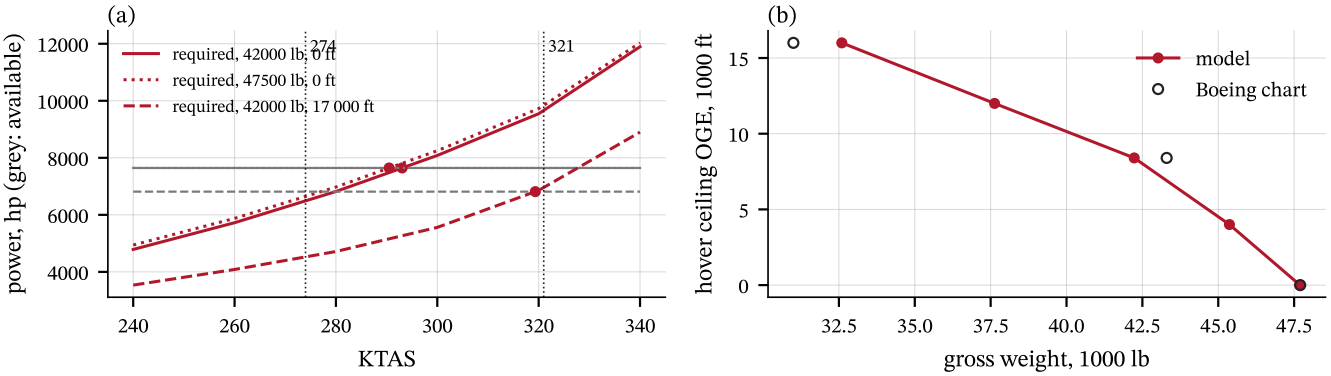


Figure 6: V-22 performance against Boeing's charts [6] (standard day, 6150 shp engines): (a) power required in airplane mode at 84 % rotor speed and power available (grey: the engines' rating or the drive's torque limit, whichever is less); the dotted verticals are the chart's maximum speeds at 42 000 lb. (b) Hover ceiling out of ground effect at 100 % rotor speed.

Table 10: V-22 performance, model and Boeing's charts.

Quantity	model	test [6]
maximum level speed, 42000 lb, 0 ft	293 KTAS	274 KTAS
maximum level speed, 47500 lb, 0 ft	291 KTAS	–
maximum level speed, 42000 lb, 17 000 ft	319 KTAS	321 KTAS
hover ceiling OGE at 0 ft	47694 lb	47 700 lb
hover ceiling OGE at 4 000 ft	45369 lb	–
hover ceiling OGE at 8 400 ft	42231 lb	43 300 lb
hover ceiling OGE at 12 000 ft	37630 lb	–
hover ceiling OGE at 16 000 ft	32595 lb	31 000 lb

Table 11: Conversion, model minus test.

Nacelle, deg	KCAS	wing lift, % of W	pitch, deg
90	0–82	−11 to −3	−3.4 to +3.0
75	37–101	−11 to +1	−2.8 to +2.9
60	60–126	−9 to −0	+0.0 to +3.9
30	110–168	−29 to −10	+2.9 to +9.0
0	116–184	−13 to −4	+0.0 to +6.0

At 42 000 lb the maximum level speed is 293 KTAS at sea level against 274, limited by the drive's torque at 84 % rotor speed, and 319 KTAS at 17 000 ft against 321, limited by the engines.

5.7 The conversion

The pitch attitude follows the flight test through the conversion within 3.3° (rms; Fig. 7, Table 11), the model nose-up of it at the higher speeds of each nacelle angle, by up to 9°: there the flaperon law's least flap (none at 30° of nacelle and 168 kt, where the test is 5° nose-down) leaves the nose higher than the test's unpublished flaps did. The wing carries 9 % of the weight less than Dunford's chart (rms), most at 30° of nacelle, where the model's wing carries 58–72 % against 83–87 %: at that angle the rotors' slipstream over the wing, which the wake model gives, decides the wing's lift, and no test checks it. The charts' weight and flaps are not published (Section 2).

5.8 Handling with the augmentation, and the start

These are not comparisons but checks of the FlightGear aircraft's augmentation (Fig. 8). In the hover one press of the stick forward gives −1.0° of pitch, held, and one to the right 1.4° of roll after a small overshoot; one of the pedals turns it at 1.6°/s and the heading holds where they are let go. With the nacelles commanded from 90 to 60° at 40 kt and nothing else touched, the aircraft accelerates to 150 kt with its pitch between −8.7 and 2.9°: the trim feed-forward takes up the trim's change, the stick's trim moving from 67 % of its travel aft of centre in the hover at 75° to 68 % forward at 120 kt at 90°, which the bare aircraft would have the pilot hold. The nacelles stop at the corridor's lower boundary until the airspeed allows 60°, and later move on past it as the upper boundary comes down. Cold, with the lever down, the engines spool up one after the other and the rotors reach their governed speed 65 s after the first engine's start.

5.9 The hover hold

The hover hold (HVR CPLD) follows the AH-64D's modern control laws [18], Bell's automatic hover hold [20] and ADS-33E's position hold and translational rate command [17], with the V-22's own way of translating: on its nacelles, the fuselage level [21], as NASA's large tiltrotor's translational rate command does [19]. Engaged at up to 40 kt over the ground, it sets the stop point, where the present velocity comes to rest at 0.08 g, and flies one profile to it, then holds it and the height (radar height below 500 ft, the barometric altitude above); the nacelles take the fore and aft acceleration, the bank the sideways, the pitch only what is past the nacelles' range. Engaged at 39 kt it stops 867 ft on, holding 25 s after; every stop ends within 1.4 ft of the point it showed, the pitch within 2.4° (Fig. 9a). In ADS-33E's check, a turn on the pedals in about 30 s, the aircraft stays within 4.5 ft of the point in still air, which meets the 10 ft circle, 6.4 ft in 20 kt of wind and 9.3 ft in 35 kt, about twice the circle's radius, its attitudes within 6.6° against the 18° allowed (Fig. 9b). The stick out of its detent commands a ground speed: 5.1 kt for a quarter of its travel, reached to 63 % in 3.8 s (2.5–5 s), overshooting by 0.2 kt, and on its release the aircraft stops and holds where it does (Fig. 9c). The hold drives the lever and the pilot can fly through it: moved for 4 s and left, the aircraft settles 7 ft lower, held within 0.1 ft (Fig. 9d). The hold's sideways loop

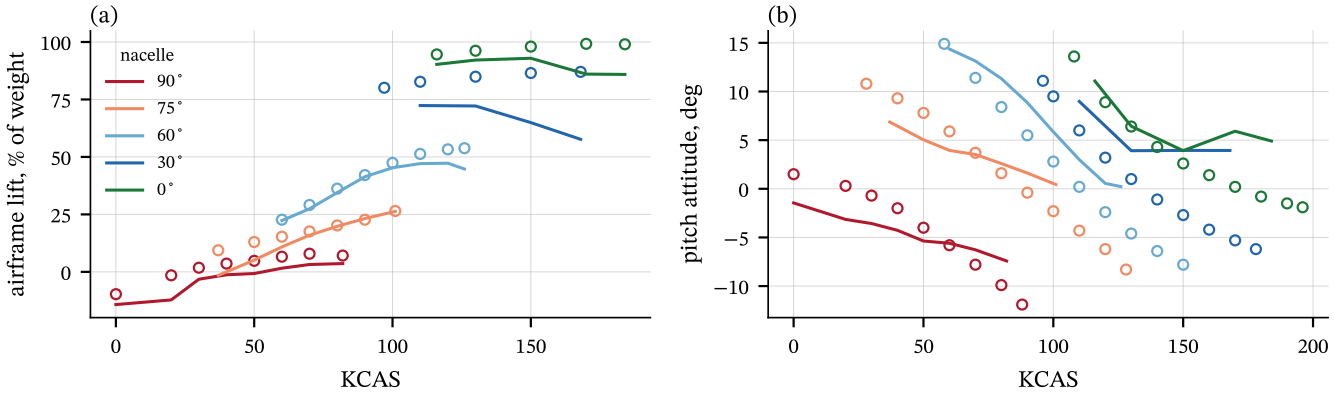


Figure 7: Trims through the conversion at 34 000 lb, sea level, against the V-22's flight test [7] figs. 7 and 9a: (a) the airframe's share of the lift, (b) pitch attitude, by nacelle angle (circles: test).

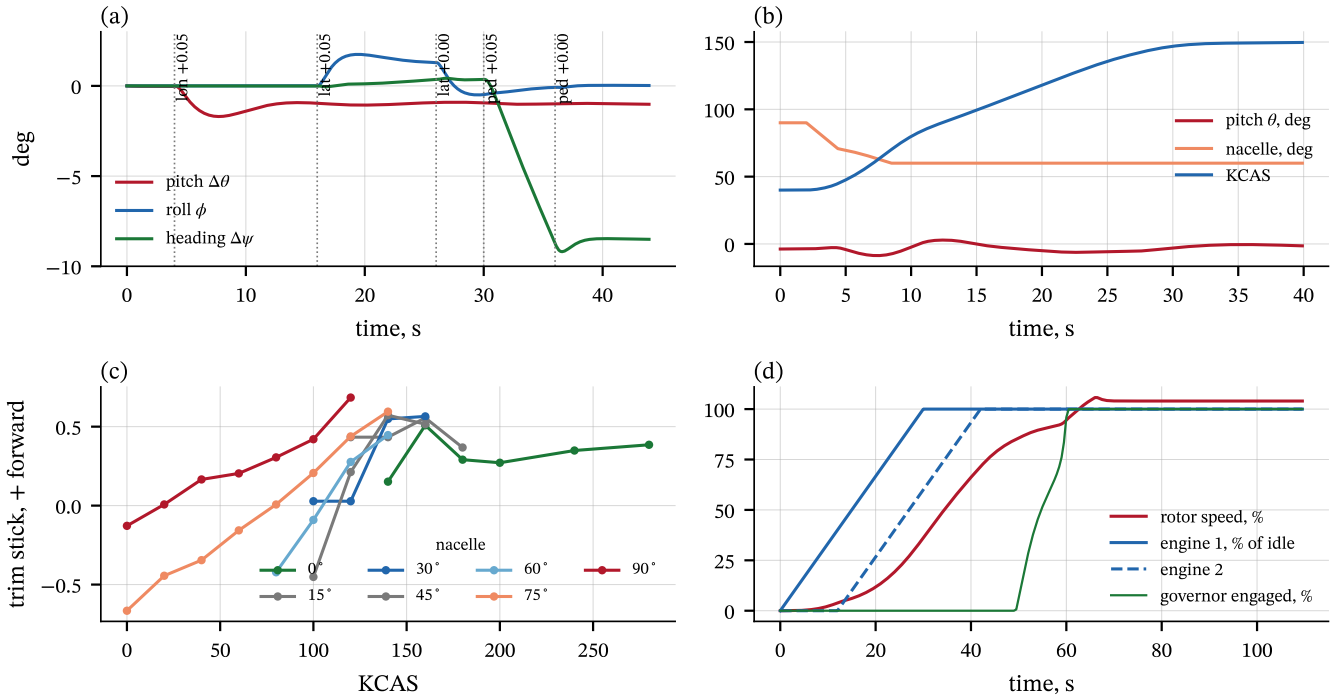


Figure 8: The FlightGear aircraft, augmentation on, reference loading: (a) single keyboard steps (5 % of travel) from the hover, stick centred; (b) a hands-off conversion from 40 kt, the nacelles commanded from 90 to 60°; (c) the trim feed-forward, the level-flight trim's stick against airspeed by nacelle angle; (d) an engine start on the ground, lever down.

Table 12: Terrain following, 200 ft set, the medium ride; after the first 5 s. Jitter: the least clearance with FlightGear's timer up to a frame (77 ms) late.

Terrain	KCAS	clearance, ft: least, mean, most			jitter	vertical g	pitch, deg
level	120	200	200	200	200	−0.02 to +0.01	+2.4 to +2.5
level	150	200	200	200	–	−0.02 to +0.03	+3.7 to +4.4
level	200	200	200	200	199	−0.02 to +0.05	+6.0 to +6.5
hills	120	198	200	202	–	−0.12 to +0.10	+0.9 to +4.6
hills	150	191	204	222	–	−0.37 to +0.20	−7.4 to +14.3
hills	200	190	330	506	–	−0.26 to +0.11	+0.7 to +13.1
ridge	120	197	255	618	197	−0.23 to +0.19	−0.5 to +5.7
ridge	150	180	270	807	–	−0.35 to +0.21	−3.7 to +14.5
ridge	200	180	294	1009	180	−0.31 to +0.20	+0.2 to +14.4
ridge	170	185	276	858	–	−0.24 to +0.21	−3.2 to +14.0

works through the roll attitude's, which needed twice its first gains; the 35 kt turn's errors come from the wind's loads turning with the aircraft faster than the hold's integrators follow.

5.10 The autopilot

The autopilot is a flight director that the augmentation flies when coupled, as the V-22's is [7]; its laws are not the V-22's, which are not public. In airplane mode the lever and the pitch share the speed and the flight path by total energy control [29]: the lever the total energy's rate, the pitch its share between climb and speed, the commanded flight path's angle fed forward. Blended by the nacelle angle from 50 to 80° it passes to the back side of the power curve, where the lever flies the climb and the pitch the speed. AUTO NAC takes the nacelle angle for the speed from the model's own level trims, the most converted angle whose trim's pitch is at most 4° (7° at 180 kt), within the corridor. The autopilot commands the augmentation's attitudes and their rates; the stick for those rates is fed forward from the airframe's measured response per inch of stick, and the augmentation damps the difference from the commanded rate rather than the rate itself, which had resisted every turn and pull-up. In pitch the airframe's first response to the stick is up to seven times its sustained rate (at

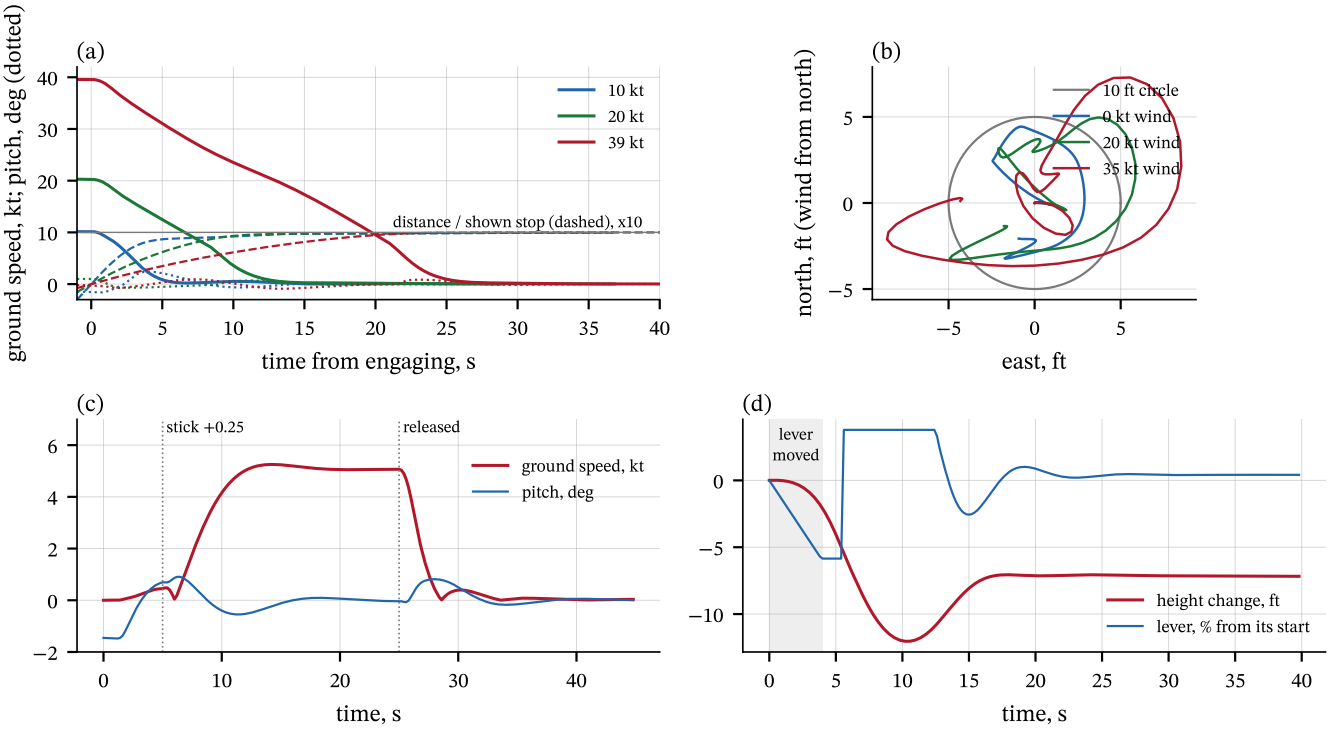


Figure 9: The hover hold, reference loading, augmentation on: (a) engaged from 10, 20 and 39 kt, the ground speed, the pitch attitude (dotted) and the distance flown over the stop it showed (dashed); (b) ADS-33E-PRF's position hold check, a 360° turn on the pedals in about 30 s in a steady wind from the north, the track about the hold point; (c) the stick out of its detent and back; (d) the lever moved down for 4 s and left.

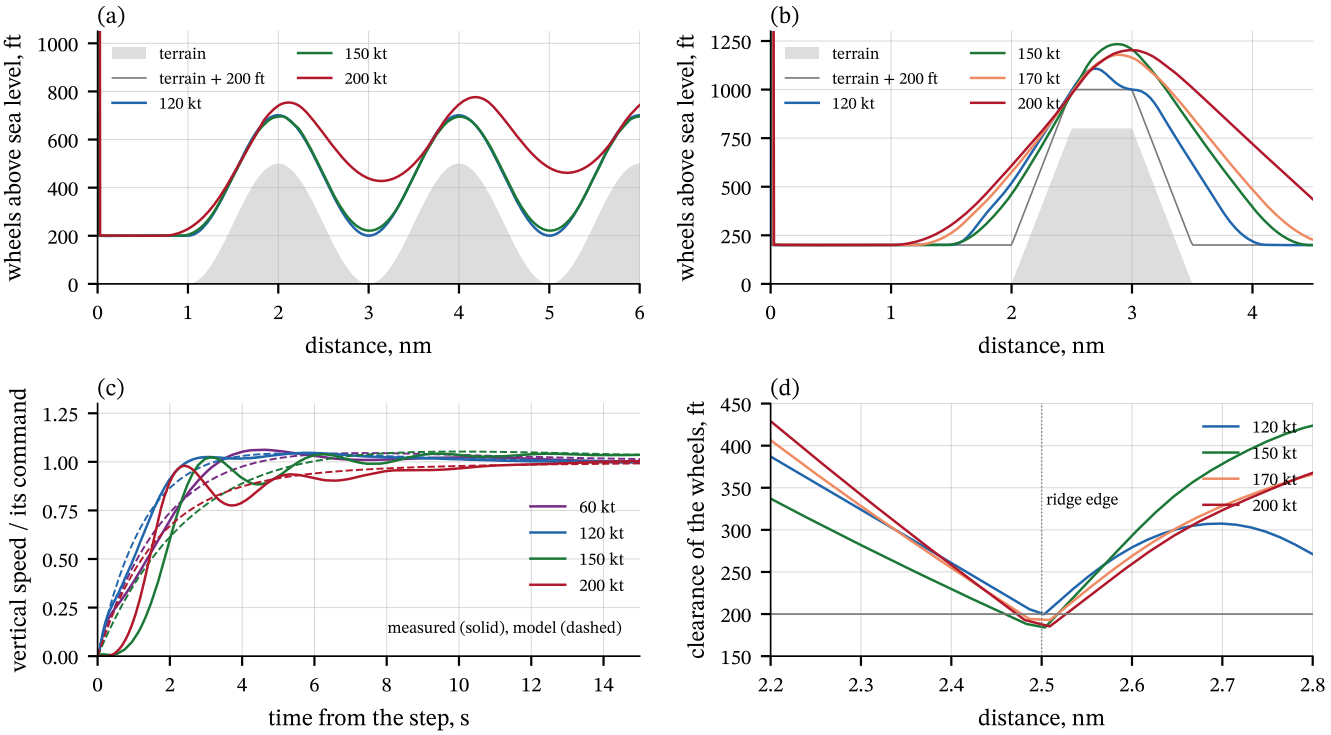


Figure 10: Terrain following at 200 ft, the medium ride (0.2 g), SPD HOLD and AUTO NAC, reference loading: (a) over hills 500 ft high and 2 nm apart; (b) over a ridge 800 ft high with 15° faces; (c) the climb's response to a step in TF's command (10 ft/s), measured, and the model the flight model keeps of it; (d) the clearance at the ridge's edge.

150 kt, 50°); the feed-forward goes through a lag and closes its error over a time both set from that ratio, which keeps the loop's gain at high frequency below one: fed straight it rang at 1 Hz and diverged at 150 kt.

Coupled at 200 kt it climbs 1000 ft without overshoot, its pitch at most 10.2°, and turns 90° banked at most 26.6° within 10 ft of its altitude. Converting from 220 to 100 kt and back under AUTO NAC it holds the altitude within 49 ft, the pitch between −3.1 and 7.5°. Engaging, the vertical speed's and the speed's references start from the aircraft's own vertical speed and acceleration and change at no more than 0.1 g, the speed's in the energy's terms, $V\dot{V}/g$ (its rate changing at no more than

$0.1\,g^2/V$, easing onto the selected speed too): stepped, the speed's rate to 2 kt/s at once and back to 0 on reaching it, an engagement at 120 kt climbing at 19 ft/s and 3 kt slow, as after FlightGear's in-air start, pitched the nose from +7.6 to −3.8° and back to +3.6°, ringing on. Now it pitches from +2.4 to −1.3° and comes back by at most 2.5°, within $0.12\,g$. An approach to a point along a course and a glide path (an ILS's, or a runway's or landing zone's, from 6 nm and 2000 ft off the line) keeps within 8 ft of the course and −26 to 27 ft of the path, slows down along it, its speed's reference stopping itself over the point (the deceleration $v^2/2d$ of its ground speed in the distance to go), and hands over to the hover hold below 35 kt over the ground at least 606 ft before the point (35 kt of airspeed in still air, 49 kt in 15 kt of headwind), running at most 0 ft past it. (Chasing the stopping profile's speed at the profile's own deceleration, the reference fell further behind as it slowed, and the hold took over at the point at 34 kt, ran up to 330 ft past it and backed up.) The hold holds the hover within 5 ft of the point or lands 0.1 ft from it, sinking at 1.7 ft/s. A go-around loses at most 3 ft. Flying a route, the speed comes down from 1.14 nm before its landing zone, where the plan drawn on the navigation display has its deceleration point at 1.33 nm, and the aircraft lands 0.1 ft from the point. Between legs it flies its own arcs, started where an arc at 80 % of the bank limit joins them, led by the roll's measured delay: past the next leg by at most 84 ft at 200 kt and 24 ft at 100 kt in 90° turns, and 33 ft in a 149° turn at 120 kt.

Terrain following (TF) holds a set clearance over the scenery's terrain. Every 0.25 s it samples the terrain along the predicted track (curving at the present turn rate) every 1.25 s of flight to 45 s ahead, on the track and 150 ft either side, and keeps the last four sweeps' points, which lie between them 0.25 s apart: one sweep's alone missed a ridge's edge by up to 1.25 s. It does not plan from the aircraft. Planned from the aircraft's state, terrain nearer than its climb's delay could be met only with everything or nothing, a relay through a delay, and the aircraft porpoised over level ground, ±8 ft/s at 200 kt. The flight model instead keeps a reference path, its vertical speed ramped at the ride's acceleration to the plan's command, and a model of the aircraft following it: two first-order lags in parallel fitted to the climb's measured response to a step, the pitch's quick share and the slower part the lever adds (Fig. 10c, rms within 0.09). The aircraft is held to the model's height with a gain set for 60° of phase margin on that measured response, 0.23–0.69/s. For a vertical speed commanded now, reached at the ride's acceleration and held to the next plan, the plan has the model's height at each point in closed form, with the most a pull-up at half the ride's acceleration then adds up to the climb the power makes, and commands the least vertical speed that clears every point. It descends into a valley and pulls up as late as half the ride allows. A point too near to clear in time, nearer than the height loop's time constant, is a height error, closed over that time with the command not eased meanwhile: FlightGear's timer, up to a frame late, otherwise left the model a fraction of a foot under the clearance and the plan climbing at full power for it. Where the lever reaches its stop and the aircraft still falls behind, the model and the reference come down to it, and the plan never counts on height the aircraft has not got. The climb the power makes comes from the power a climb takes, $P = P_{\text{level}} + b\dot{h} + c\dot{h}^2$ (b, c from three trims at each speed, the level flight's power from the present power and the energy's rate), divided by the share of a climb at constant calibrated airspeed that goes into the true airspeed's rise (5 % at 200 kt). These tables are read at the speed AUTO NAC's schedule gives the present nacelle angle: through the conversion the nacelles change the climb more than a few knots do, 2787 and 2326 ft/min at full power at 150 and 160 kt. With SPD HOLD the speed is free within 20 kt of its reference. Below half the clearance, or without terrain data, it climbs at full power.

Table 13: Steady descents: where the trim collective is least and greatest against the descent rate, the rotors' mean V_z/v_h there on their own scale, Johnson's N and X at their V_x/v_h, and the last descent rate that trims (ft/min).

build	kt	collective least: ft/min, z, z_N			greatest: ft/min, z, z_X			last
V-22	0	1933	−0.43	−0.45	7147	−1.47	−1.47	7759
	10	2054	−0.44	−0.46	6605	−1.34	−1.35	7303
	20	2362	−0.51	−0.48	5279	−1.09	−1.22	6846
	30	2689	−0.57	−0.54	4479	−0.95	−1.05	6162
	40	–	–	–	–	–	–	5705
	50	–	–	–	–	–	–	4564
	60	–	–	–	–	–	–	3880
XV-15	0	1596	−0.44	−0.45	5229	−1.40	−1.47	5984
	10	1756	−0.47	−0.46	4701	−1.22	−1.33	5259
	20	2191	−0.58	−0.51	3555	−0.94	−1.12	5259
	30	–	–	–	–	–	–	4171
	40	–	–	–	–	–	–	3808
	50	–	–	–	–	–	–	3445
	60	–	–	–	–	–	–	3083

Table 12 and Fig. 10 give TF over level ground, hills 500 ft high and 2 nm apart, and a ridge 800 ft high whose 15° faces are steeper than any climb of the aircraft's. Over level ground the pitch holds within 0.01° (rms), 0.01° with the timer's jitter. Over the hills at 120 kt the clearance stays within 198–202 ft; at 200 kt the valleys are narrower than the 2000 ft/min descent and the ride allow the aircraft to follow, and it stays up to 506 ft over them (mean 330). The ridge leaves at least 180 ft at 120 to 200 kt (185 ft at 170 kt), its edge flown at the climb's limit: at 150 kt, the nacelles at 30°, the lever near its stop. The vertical acceleration stays within $0.37\,g$, the ride's $0.2\,g$ with what the height loop adds, and no run flies up. The timer's jitter changes the least clearance by at most 1 ft. The plan's 45 s cover 1000 ft of rise with the pull-up at the least full-power climb TF meets at 120 to 200 kt (1913 ft/min, at 120 kt); higher or nearer terrain, or terrain coming into view round a turn, can be cleared only as the fly-up allows.

5.11 Descent through the vortex ring state

Johnson's algorithm (his table 3 [30]) is coded as he gives it but for two misprints, read as his text has them: the last segment's far end is matched without its slope at E, not D, and the cubics' slope is $b + 2cV_z + 3dV_z^2$. The increments of the inflow at N and X are his table's values less axial momentum theory's at V_{zN} and V_{zX}, faded with V_x: so his figures 38 and 39 come out within the scan's line width (at $V_x/v_h = 0.8$ the least $V_z + v$ is 0.537, the figure's 0.53; with momentum theory taken at the moved point and V_x it would be 0.78).

Trimmed in steady descents in helicopter mode, from level flight down in steps of $0.05\,v_h$ at 0 to 60 kt (370 trims of the two builds), each rotor's uniform inflow is Johnson's at its own axial and edgewise velocity within $0.0010\,v_h$ on average (Fig. 11a, b): within 0.04 where the rotor stays in or out of

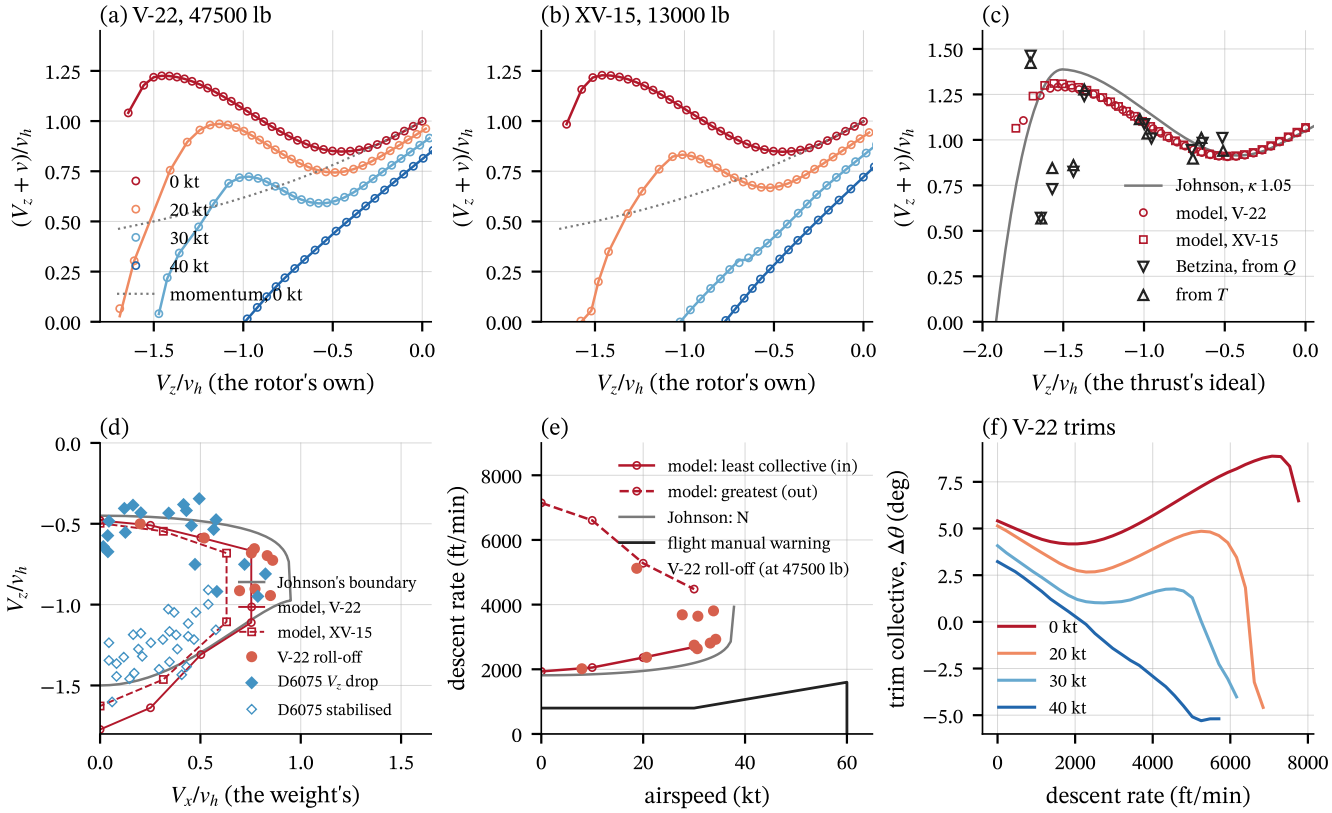


Figure 11: Steady descents in helicopter mode, 1000 ft, out of ground effect, still air, the reference loadings. (a, b) Each rotor's total flow through the disc against its axial velocity, on its own wake's v_h: the trims (circles) and Johnson's model at the same axial and edgewise velocities (lines). (c) Vertical descent on the thrust's ideal v_h, against Betzina's tiltrotor rotor in a wind tunnel, the inflow from its torque and its thrust [31], and Johnson's curve as he draws it against them [30]. (d) The trims' boundary, where their collective is least and greatest against the descent rate, on the weight's v_h as the flight tests are [30, 32]. (e) The V-22's boundary in the flight manual's terms, its warning's boundary, Johnson's on the weight's v_h and the flight tests' roll-offs at this weight. (f) The V-22's trim collective.

his region through the revolution, the most past X, where the tables are a few hundredths off, and up to 0.10 where it sits on the region's edge at 0.95 v_h edgewise (at 40 kt): there its velocities over v_h, which ripple with the blades' passage, cross the edge within each revolution, where his increment, whose slopes at N and X do not fade with it, is still a tenth of v_h. Descending vertically the V-22 first needs less power, as momentum theory has it, and then more: 45 % more than in the hover at Johnson's X. Its trim collective, which at a fixed lever decides the heave mode's stability, is least at 1933 ft/min and greatest at 7147 ft/min (Table 13, Fig. 11f); between them a descent at a fixed lever runs away. On the rotors' own scale these are at $V_z/v_h = -0.43$ and -1.47, against Johnson's N and X at -0.45 and -1.5, the difference the attitude, the thrust and the edgewise velocity changing along each descent; the XV-15's at -0.44 and -1.40. With forward speed the unstable band narrows and closes as Johnson's boundary does. Past X the total flow falls steeply into the turbulent wake state, and the trims stop converging there, past 7759 ft/min (V_z/v_h -1.64) in vertical descent: the thrust turns back on the collective.

Johnson scales the flight tests he gathered by the aircraft's weight, his model by the rotor's ideal v_h. On the weight's scale the V-22's boundary in vertical descent is at $V_z/v_h = -0.48$, against his -0.45 (Fig. 11d): the wake's own scale is 1.065 times the thrust's ideal, and the rotors carry the wing's download besides the weight. The V-22's roll-offs in flight test [32] lie past the model's boundary, but one at $V_x/v_h = 0.2$, which lies on it. In the flight manual's terms (Fig. 11e) the model's boundary at 47 500 lb is at 1933 ft/min in the hover, rising with the airspeed; the manual's warning, 800 ft/min to 30 KCAS, sits below it with a margin. On the thrust's ideal scale (Fig. 11c) the vertical descents follow Johnson's curve as he draws it against Betzina's tiltrotor rotor; his curve passes within 0.08 (rms) of its inflow between N and X, and near X, where the test's points jump between two levels, up to 0.7 from them.

Flown through Johnson's V-22 encounter [30] (his table 6, 46 100 lb; $V_x/v_h = 0.6$, $V_z/v_h = -0.35$ on his v_h of 65.2 ft/s), the aircraft follows his calculation for the first seconds, V_z/v_h -0.42 at 2 s against his -0.42, and then falls behind, -0.60 at 10 s against -0.79 (Fig. 12). Its rotors start at $V_z/v_h = -0.28$ on their own scale, not -0.35: the download, the wake's induced losses and the shaft's tilt take a fifth off, so N lies deeper. Its descent runs away from about 20 s, reaching -1.25 at 30 s, the normal acceleration down to 0.83 g at 29 s (his 0.75 g at 12 s). Johnson's aircraft, its two rotors identical, dropped and did not roll; nor did his with sideslip or the centre of gravity offset, and he needed rotors made different to calculate a roll-off. This one drops and then, at the bottom of the drop with its rotors at $V_z/v_h = -1.56$, rolls off out of the trim's last hundredths of a degree of asymmetry: a yaw rate slows one rotor edgewise and speeds the other; the slower goes deeper into the vortex ring, its inflow and power rise and its thrust falls, which rolls the aircraft towards it and yaws it further by the rotor's torque. The augmentation's differential collective reaches its limit, 1.5°, at 42 s, and the roll passes 10° at 42 s, 45° at most. Trimmed in 15° of sideslip it drops

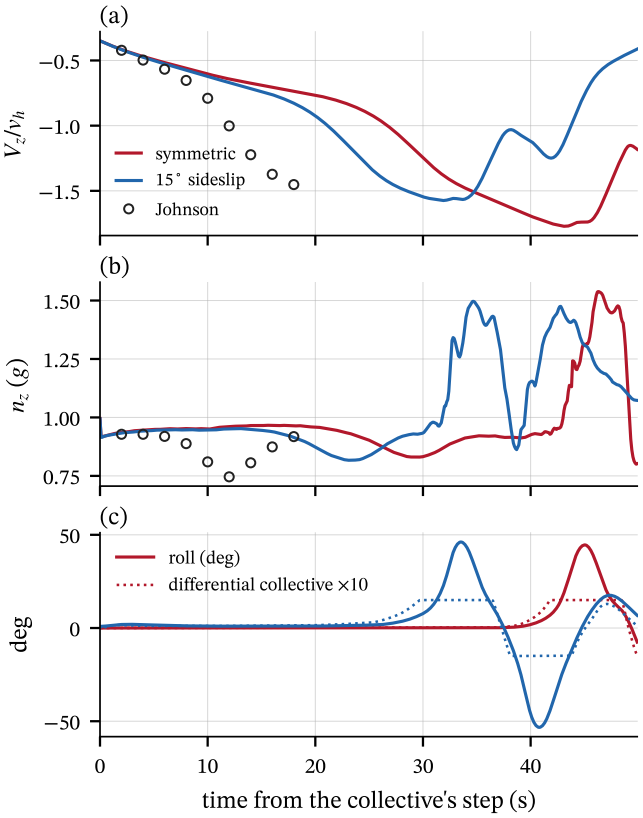


Figure 12: Johnson's V-22 encounter [30]: from a descent at 26.8 kt on a −30° path at 46 100 lb, his collective, 1° down at once and another degree over 10 s; the rotor speed held, the augmentation holding the attitude. (a) The vertical velocity on his v_h, 65.2 ft/s, symmetric and trimmed in 15° of sideslip, and his symmetric calculation (traced); (b) the normal acceleration; (c) the roll and the augmentation's differential collective.

faster and rolls off at 31 s. The roll-off comes from each rotor's own velocities, but at the bottom of the drop rather than on entering it, where the V-22's flight tests met it (Fig. 11d); the model's wakes do not act on each other, nor is the vortex ring's unsteadiness modelled, both of which Johnson names as possible causes [30].

Outside the vortex ring state nothing changed: with the model in place, all 47 runs behind Sections 5.1 to 5.10, the hover, the level-flight power, the frequency responses, the download, the performance, the conversion, the handling, the hover hold, the autopilot and terrain following, give the same results to the last digit; none enters Johnson's region.

6 Corrections found during validation

The changes that the comparisons forced are listed here because each removed a physical or numerical inconsistency rather than tuning a coefficient.

1. *The sections' stall.* The sections first had C-81's NACA 0012 stall, the lift falling to a flat plate's within 6°. Its negative lift slope past the maximum took away the gimbal's aerodynamic damping, and at high thrust the gimbal diverged. The 64-series sections stall from the trailing edge and fall gently [12]; so do the model's now, the shape fitted to the hover test.
2. *The nacelles' mass.* Each nacelle's centre of gravity was placed on the wrong side of the conversion axis, which put the aircraft's 5 in forward.
3. *The governor.* Its error had the wrong sign: a rotor slower than its reference took more collective, so it slowed further. And its engagement was gated at an absolute rotor speed of 85–95 %, which turned it off in airplane mode at 84 %; it is gated relative to the reference now.
4. *The flaperons in slow airplane mode.* Retracted with the nacelles down, they left the wing unable to carry the aircraft at the corridor's lowest airplane-mode speeds; the schedule keeps 40° there.
5. *Test weights.* The tests' weight changes went into the cargo's point mass, which moved the centre of gravity with them; they are at the design centre of gravity now.
6. *The drive's torque limit* was applied as a power proportional to the rotor speed, so a start had almost no torque to turn the rotors up; it acts as a torque now, and the flight idle is the power the rotors need at flat pitch at their governed speed.
7. *The gear in the comparisons.* The harness first ran with the gear extended, JSBSim's default. Its drag stood in for drag the airframe lacks: the V-22's fitted drag (then 4.7 ft^2) met Boeing's polar and the airplane-mode speeds only with the gear down, and the XV-15's airplane-mode power looked close. All comparisons in forward flight are now gear up, the V-22's drag refitted, and the XV-15's shortfall reported (Section 5.2).
8. *The nacelle rate.* The nacelles' tilt rate, which moves the hubs and adds to the shafts' pitch rate in the flapping, was taken after the step had already set the new nacelle angle, so it was zero throughout. Taken from the step itself it changes no trim, and the pitch attitude through the conversion run (Section 5.8) by up to 1.6°.
9. *The handling runs' start.* The harness let the aircraft go one frame after setting the lever and centring the stick, before the controls' chain (the mixing, its actuators, the governor) had carried them through, and set the augmentation's integrators from that frame's values: the hover run began with a 2 % sag in rotor speed, drifted 3 kt and lost 40 ft that the aircraft does not. It now waits a whole period of the controls' channels (four frames) before letting go, and the hover holds its rotor speed and height from the start.
10. *The trim feed-forward in rearward flight.* The augmentation's trim feed-forward looked its stick up against the airspeed's magnitude, so that flying backwards it gave the forward stick of forward flight: in FlightGear the aircraft let go in the hover drifted aft, pitched nose down and diverged with the augmentation's integrator at its limit. It takes the forward airspeed now (the hover's stick in rearward and sideward flight), carried past 90° of nacelle along its slope from 75 to 90.
11. *The trim feed-forward and the flaperons.* Taken from level trims against the nacelle angle and the airspeed alone, it held the stick for the flaperons those trims had. Once the flaperons came from the flight state, the automatic landing with both engines out lowered them to 40° where level flight has few: the stick's trim was a quarter of its travel off, the augmentation's half of it could not hold the nose as the nacelles came up, and it rose to 12° with 2° down commanded. The trims are now taken with the flaperons held at 0, 20, 40 and 67°, and the feed-forward is looked up at the flaperons there are.

Table 14: Summary: ✓ within the test's scatter, ~ partly, × not reproduced. Fitted: the comparison checks the shape only.

XV-15 rotor: hover power, C_T/σ = 0.08–0.18 (stall fitted)	✓
XV-15 rotor: hover power at low thrust; collective gradient	~
XV-15 download against flaps (fitted)	✓
XV-15 power through the conversion (over at both altitudes)	×
XV-15 power in helicopter mode, 45–94 kt	×
XV-15 power in airplane mode, gear up (Ferguson's added drag)	×
XV-15 trim elevator in airplane mode	✓
XV-15 trim elevator through the conversion and in helicopter mode	×
V-22 airframe polar (drag fitted)	✓
V-22 maximum level speed, sea level and 17 000 ft	✓
V-22 hover ceiling (sea level fitted)	✓
V-22 download at 67°	~
V-22 pitch attitude through the conversion	✓
V-22 wing lift share through the conversion	~
XV-15 cruise roll and sideslip frequency responses	✓
XV-15 cruise pitch frequency responses (stiffness 2.9×)	×
XV-15 hover pitch frequency response	✓
XV-15 hover yaw, roll and heave frequency responses	~
Steady descents: each rotor's inflow, Johnson's VRS model	✓
VRS boundary against the flight tests (weight's v_h)	~
Johnson's V-22 encounter: V_z drop (later), roll-off (deeper)	~

12. *Yaw at low speed.* The augmentation only damped the yaw rate, through a washout: a steady yaw moment, a tailwind on the fins, turned the aircraft at will, and the pedals, opposed by the damper, were weak (a keyboard step turned it 0.6°). Below 50 kt the pedals now command a yaw rate, held by an integrator with the pedals' full travel.
13. *The elevator trim.* The original aircraft's controls file set FlightGear's elevator trim to 0.25 at the start, a YASim setting that the augmentation took as a quarter of the stick held forward: coupled, the flight director's pitch cue sat 5° above the aircraft. It starts at 0 now.
14. *The XV-15's wing incidence.* Bell's wing tables are against the fuselage's angle of attack, the wing's 3° of incidence in them (Bell's $\alpha_W = \alpha_F - \Delta\alpha_{R/W}$ [1]); the XV-15 build added the incidence again, so its wing, and the downwash it sheds on the tail, saw 3° more than the aircraft's: at the frequency responses' cruise it trimmed 2.5° nose-down of where it does now. The V-22's build had none added.
15. *The XV-15's tail, ailerons and inertias.* Bell's 1973 data made the XV-15 two to three times as stiff and as sensitive in pitch as its flight tests' frequency responses [27]. The flight tests had shown Ferguson that the data were out of date: the inertias without the aircraft's modifications and instrumentation, the controls' effectiveness and the tails' dynamic pressure questionable [26]. The model takes his revisions (Section 3), and the V-22's tail and ailerons, which are the XV-15's coefficients, with them.

7 Assessment

Table 14 summarises. The model reproduces what defines the tiltrotor's performance: the proprotor's hover power at operating thrust, the wing's download against its flaps, the airframe's polar, and the V-22's speeds and hover ceilings, the last three after one fitted drag area and one fitted torque limit, both against Boeing's own charts. Through the conversion the V-22's pitch attitude follows its flight test. Its open items are the helicopter mode's power against speed, where the XV-15's test shows a shallower bucket than the blade elements and the wake give; the drag Bell's coefficients lack, which the V-22 build supplies by a fit and the XV-15 build by Ferguson's flight-test terms, a fifth short in airplane mode; the conversion's excess power and missing nose-up moment, both pointing to the rotors' wakes over the wing; the wing's share of the lift at intermediate nacelle angles; and the V-22 wing's download. None of the V-22's dynamic responses could be compared: no frequency responses or time histories of the aircraft are public. The XV-15's are [27]: with Ferguson's revised data its cruise roll and sideslip and its hover pitch follow them, its hover yaw nearly, while in cruise it is too stiff in pitch, by its stabiliser or its pitch inertia, which those tests cannot separate. The V-22 shares the XV-15's tail and wing coefficients but not its inertias. Its augmentation is a plausible control law, not the V-22's. In descent each rotor's inflow is Johnson's vortex ring state model, which puts the instability where the flight tests met it on the rotor's own scale; on the weight's, as the flight tests are scaled, the V-22's wing download and the wake's induced losses put it at $V_z/v_h = -0.48$ in vertical descent against their −0.45, and its roll-off comes at the bottom of the drop, not on entry.

8 Conclusions

- A blade-element model of the gimballed tiltrotor, both rotors blade by blade with their gimbals, the blades' torsion, pitch–flap coupling and finite-state wakes, the wing in their ring-vortex wakes, the interconnected drive and the conversion corridor, runs in plain JSBSim, generated from data. It is the flight model of the FlightGear V-22.
- Built first as the XV-15 from Bell's simulation data and NASA's tests, its rotor gives the full-scale rotor's hover power at operating thrust, and its download follows the flight test's with the flap.
- Carried to the V-22 with its published data, one fitted drag area and one fitted torque limit, it gives Boeing's airframe polar, maximum speeds within 15 kt and hover ceilings within 5 %, and the flight test's pitch attitude through the conversion.
- Against the XV-15's frequency responses from flight, Bell's 1973 tail and inertias made it far too stiff and sensitive in pitch; with Ferguson's revisions of them its cruise roll and sideslip and its hover pitch follow the flight tests, and its cruise pitch remains too stiff.
- Bell's airframe coefficients miss part of the flying aircraft's drag; with the spinners' and the pylons' drag Ferguson added from the flight tests, the XV-15 build still needs a fifth less power than its flight test in airplane mode, and through the conversion it needs more: the rotors' wakes over the wing there, which no test checks, are the model's weakest part.
- The helicopter mode's power against speed, the wing's lift through the middle of the conversion and the V-22's download are documented, not tuned away.
- In descent each rotor's inflow follows Johnson's vortex ring state model on its own axial and edgewise velocity,

joined to the wake's momentum theory without changing any other result: descending vertically at a fixed lever the V-22 runs away between 1933 and 7147 ft/min, and in Johnson's encounter it drops and then rolls off, out of the two rotors' own velocities.

- Its autopilot's terrain following plans a reference path over the terrain ahead and holds the aircraft to its measured response: over hills and a ridge steeper than its climb it keeps within 10 % of a 200 ft clearance at 120 to 200 kt, without porpoising over level ground.

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